

DIGT 3107 - Room/Hotel Booking System

Phase 2

Student#:	219941855	220329405	220486114
Sur (Family) Name:	Carter	Chawda	Dhawan
Given Name:	Toby	Gira	Komal
Section:	A	A	A

Phase 2: ERD to SQL Implementation

Tables:

1. **Guests:** Stores information about all the guests who book rooms at the university guest house. Key features of this table include: The contact information can be either a phone number or email, it tracks the staff status for discount eligibility, and it has an autoincrementing primary key. The test data is inspired by friends' characters and shows different cases where some may be staff.
2. **Payment Methods:** Defines the available payment methods to guests. It is a lookup table for payment types and can be expanded for different types in the future. The data we've provided are the top three common types of payment methods: credit card, debit card, and cash.
3. **Room Types:** This table defines the three room categories available at the University Guest House, the capacity of each room and the pricing. This table helps manage pricing for each room, and defines guest capacity for each room. The data provided defines the three types of rooms available at the university guest house.
4. **Rooms:** This table represents the physical rooms in the guest house, each room has a unique identifier and more rooms can be added easily in the future should there be new construction. The data shows a total of 10 rooms and their room types.
5. **Bookings:** This table records all reservations including active, completed, and cancelled bookings with timestamps. It links guest, room and payment information together while also recording whether staff discount would apply. The data shows a series of bookings made by different guests, the status of their bookings, when they were made, check in and check out and whether they are receiving a staff discount on the booking.

Table Structures:

Guests Table:

	Field	Type	Null	Key	Default	Extra
►	guest_id	int	NO	PRI	NULL	auto_increment
	name	varchar(100)	NO		NULL	
	contact_info	varchar(254)	NO		NULL	
	is_staff	tinyint(1)	NO		0	

	Field	Type	Null	Key	Default	Extra
▶	guest_id	int	NO	PRI	NULL	auto_increment
	name	varchar(100)	NO		NULL	
	contact_info	varchar(254)	NO		NULL	
	is_staff	tinyint(1)	NO		0	

Payment Methods Table:

	Field	Type	Null	Key	Default	Extra
▶	payment_method_id	int	NO	PRI	NULL	auto_increment
	method_type	varchar(11)	NO		NULL	

Room Types Table:

	Field	Type	Null	Key	Default	Extra
▶	room_type_id	int	NO	PRI	NULL	auto_increment
	type_name	varchar(12)	NO		NULL	
	capacity	int	NO		NULL	
	nightly_price	decimal(5,2)	NO		NULL	

Rooms Table:

	Field	Type	Null	Key	Default	Extra
▶	room_id	int	NO	PRI	NULL	auto_increment
	room_number	varchar(3)	NO	UNI	NULL	
	room_type_id	int	NO	MUL	NULL	

Bookings Table:

	Field	Type	Null	Key	Default	Extra
▶	booking_id	int	NO	PRI	NULL	auto_increment
	guest_id	int	NO	MUL	NULL	
	room_id	int	NO	MUL	NULL	
	payment_method_id	int	NO	MUL	NULL	
	booking_datetime	datetime	NO		NULL	
	check_in	datetime	NO		NULL	
	check_out	datetime	NO		NULL	
	is_cancelled	tinyint(1)	NO		0	
	discount_applied	tinyint(1)	NO		0	

Sample Data:

Guests Sample Data:

	guest_id	name	contact_info	is_staff
▶	1	Rachel Green	rachel.green@gmail.com	0
	2	Monica Geller	monica.geller@hotmail.com	1
	3	Phoebe Buffay	212-555-0123	0
	4	Joey Tribbiani	joey.tribbiani@yahoo.com	0
	5	Chandler Bing	212-555-0198	1
	6	Ross Geller	ross.geller@outlook.com	1
	7	Janice Hosenstein	212-555-0156	0
	8	Gunther	gunther.centralperk@gmail.com	1
	9	Mike Hannigan	212-555-0187	0
	10	Emily Waltham	emily.waltham@hotmail.com	0
*	NULL	NULL	NULL	NULL

Payment Methods Sample Data:

	payment_method_id	method_type
▶	1	Credit Card
	2	Debit Card
	3	Cash
*	NULL	NULL

Room Types Sample Data:

	room_type_id	type_name	capacity	nightly_price
▶	1	Single	1	100.00
	2	Duplex	2	250.00
	3	Presidential	4	500.00
*	NULL	NULL	NULL	NULL

Rooms Sample Data:

	room_id	room_number	room_type_id
▶	1	101	1
	2	102	1
	3	103	1
	4	104	1
	5	105	1
	6	201	2
	7	202	2
	8	203	2
	9	204	2
	10	301	3
*	NULL	NULL	NULL

Bookings Sample Data:

[illegible]